ADI-FDTD for Inhomogeneous Media

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1 Preliminaries

1.1 Maxwell's Equations

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} \tag{1.1a}$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \tag{1.1b}$$

Expanding (1.1a) into components:

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \varepsilon \frac{\partial E_x}{\partial t} + \sigma E_x \tag{1.2a}$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = \varepsilon \frac{\partial E_y}{\partial t} + \sigma E_y \tag{1.2b}$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \varepsilon \frac{\partial E_z}{\partial t} + \sigma E_z \tag{1.2c}$$

Expanding (1.1b) into components:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu}\left(\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y}\right) \tag{1.3a}$$

$$\frac{\partial H_y}{\partial t} = \frac{1}{\mu}\left(\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z}\right) \tag{1.3b}$$

$$\frac{\partial H_z}{\partial t} = \frac{1}{\mu}\left(\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x}\right) \tag{1.3c}$$

1.2 Yee Grid in AMReX

The Yee grid stores different field components at different physical locations within each computational cell. In AMReX this staggering is represented by the `IndexType` used to define each `MultiFab`: a cell-centered direction has index type 0 and is located at $i + \frac{1}{2}$, while a nodal direction has index type 1 and is located at i. The implementation uses

$$\mathbf{E}_\alpha : \text{cell-centered in the } \alpha \text{ direction and nodal in the other two directions,}$$
$$\mathbf{H}_\alpha : \text{nodal in the } \alpha \text{ direction and cell-centered in the other two directions.}$$

Thus the component locations are

Component	x location	y location	z location
$E_{x,i,j,k}$	$i+\frac{1}{2}$	j	k
$E_{y,i,j,k}$	i	$j+\frac{1}{2}$	k
$E_{z,i,j,k}$	i	j	$k+\frac{1}{2}$
$H_{x,i,j,k}$	i	$j+\frac{1}{2}$	$k+\frac{1}{2}$
$H_{y,i,j,k}$	$i+\frac{1}{2}$	j	$k+\frac{1}{2}$
$H_{z,i,j,k}$	$i+\frac{1}{2}$	$j+\frac{1}{2}$	k

The half-cell offsets therefore do not appear explicitly as array indices such as $i+\frac{1}{2}$. They are encoded in the `MultiFab` staggering. The integer offsets in the finite-difference stencil choose the two source values that bracket the field being updated. For example, $H_{x,i,j,k}$ is located at $j+\frac{1}{2}$ in the y direction, while $E_{z,i,j,k}$ and $E_{z,i,j+1,k}$ are located at j and $j+1$. Therefore

$$\left.\frac{\partial E_z}{\partial y}\right|_{H_{x,i,j,k}} \approx \frac{E_{z,i,j+1,k} - E_{z,i,j,k}}{\Delta y}. \tag{1.4}$$

Similarly, $E_{x,i,j,k}$ is located at j, while $H_{z,i,j-1,k}$ and $H_{z,i,j,k}$ are located at $j-\frac{1}{2}$ and $j+\frac{1}{2}$, so

$$\left.\frac{\partial H_z}{\partial y}\right|_{E_{x,i,j,k}} \approx \frac{H_{z,i,j,k} - H_{z,i,j-1,k}}{\Delta y}. \tag{1.5}$$

This is why the magnetic update below uses forward differences such as $j+1$ and $k+1$, while the electric update uses backward differences such as $j-1$ and $k-1$: each derivative is centered at the staggered location of the field component being updated.

2 FDTD Method

The explicit Yee FDTD update used here is written in three stages, matching the implementation: first update $\mathbf{H}$ by a half step, then update $\mathbf{E}$ by a full step, and finally update $\mathbf{H}$ by another half step. Define the update coefficients

$$C_a = \frac{2\varepsilon - \sigma\Delta t}{2\varepsilon + \sigma\Delta t}, \qquad C_b = \frac{2\Delta t}{2\varepsilon + \sigma\Delta t}, \qquad D_b = \frac{\Delta t}{\mu}, \tag{2.1}$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity.

2.1 First Magnetic Half-Step

$$H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{2}\left[\frac{E^n_{z,i,j+1,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}}{\Delta y} - \frac{E^n_{y,i,j+\frac{1}{2},k+1} - E^n_{y,i,j+\frac{1}{2},k}}{\Delta z}\right] \tag{2.2a}$$

$$H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{2}\left[\frac{E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta z} - \frac{E^n_{z,i+1,j,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}}{\Delta x}\right] \tag{2.2b}$$

$$H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{2}\left[\frac{E^n_{y,i+1,j+\frac{1}{2},k} - E^n_{y,i,j+\frac{1}{2},k}}{\Delta x} - \frac{E^n_{x,i+\frac{1}{2},j+1,k} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta y}\right]. \tag{2.2c}$$

In the AMReX Yee grid, with vacuum media, the magnetic field updates are implemented as

$$B_x(i,j,k) = B_x(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_z(i,j+1,k) - E_z(i,j,k)}{\Delta y} - \frac{E_y(i,j,k+1) - E_y(i,j,k)}{\Delta z}\right] \tag{2.3a}$$

$$B_y(i,j,k) = B_y(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_x(i,j,k+1) - E_x(i,j,k)}{\Delta z} - \frac{E_z(i+1,j,k) - E_z(i,j,k)}{\Delta x}\right] \tag{2.3b}$$

$$B_z(i,j,k) = B_z(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_y(i+1,j,k) - E_y(i,j,k)}{\Delta x} - \frac{E_x(i,j+1,k) - E_x(i,j,k)}{\Delta y}\right]. \tag{2.3c}$$

2.2 Electric Full-Step

$$E^{n+1}_{x,i+\frac{1}{2},j,k} = C^{n+1}_{a,i+\frac{1}{2},j,k} E^n_{x,i+\frac{1}{2},j,k} + C^{n+1}_{b,i+\frac{1}{2},j,k}\left[\frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}\right] \tag{2.4a}$$

$$E^{n+1}_{y,i,j+\frac{1}{2},k} = C^{n+1}_{a,i,j+\frac{1}{2},k} E^n_{y,i,j+\frac{1}{2},k} + C^{n+1}_{b,i,j+\frac{1}{2},k}\left[\frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}\right] \tag{2.4b}$$

$$E^{n+1}_{z,i,j,k+\frac{1}{2}} = C^{n+1}_{a,i,j,k+\frac{1}{2}} E^n_{z,i,j,k+\frac{1}{2}} + C^{n+1}_{b,i,j,k+\frac{1}{2}}\left[\frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}\right]. \tag{2.4c}$$

In the AMReX Yee grid, with vacuum media, the electric field updates are implemented as

$$E_x(i,j,k) = E_x(i,j,k) + c^2\Delta t \left[\frac{B_z(i,j,k) - B_z(i,j-1,k)}{\Delta y} - \frac{B_y(i,j,k) - B_y(i,j,k-1)}{\Delta z}\right] \tag{2.5a}$$

$$E_y(i,j,k) = E_y(i,j,k) + c^2\Delta t \left[\frac{B_x(i,j,k) - B_x(i,j,k-1)}{\Delta z} - \frac{B_z(i,j,k) - B_z(i-1,j,k)}{\Delta x}\right] \tag{2.5b}$$

$$E_z(i,j,k) = E_z(i,j,k) + c^2\Delta t \left[\frac{B_y(i,j,k) - B_y(i-1,j,k)}{\Delta x} - \frac{B_x(i,j,k) - B_x(i,j-1,k)}{\Delta y}\right]. \tag{2.5c}$$

2.3 Second Magnetic Half-Step

$$H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{2}\left[\frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y} - \frac{E^{n+1}_{y,i,j+\frac{1}{2},k+1} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta z}\right] \tag{2.6a}$$

$$H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{2}\left[\frac{E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta z} - \frac{E^{n+1}_{z,i+1,j,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta x}\right] \tag{2.6b}$$

$$H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{2}\left[\frac{E^{n+1}_{y,i+1,j+\frac{1}{2},k} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j+1,k} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta y}\right]. \tag{2.6c}$$

In the AMReX Yee grid, with vacuum media, the second magnetic field updates are implemented as

$$B_x(i,j,k) = B_x(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_z(i,j+1,k) - E_z(i,j,k)}{\Delta y} - \frac{E_y(i,j,k+1) - E_y(i,j,k)}{\Delta z}\right] \tag{2.7a}$$

$$B_y(i,j,k) = B_y(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_x(i,j,k+1) - E_x(i,j,k)}{\Delta z} - \frac{E_z(i+1,j,k) - E_z(i,j,k)}{\Delta x}\right] \tag{2.7b}$$

$$B_z(i,j,k) = B_z(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_y(i+1,j,k) - E_y(i,j,k)}{\Delta x} - \frac{E_x(i,j+1,k) - E_x(i,j,k)}{\Delta y}\right]. \tag{2.7c}$$

This explicit method is simple and local, but the time step is restricted by the CFL stability condition. For a uniform lossless grid, a typical stability limit is

$$\Delta t \le \frac{1}{c\sqrt{\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2}}}, \qquad c = \frac{1}{\sqrt{\mu\varepsilon}}. \tag{2.8}$$

3 ADI Method

The ADI-FDTD update advances the fields by a full time step Δt in two half steps of size $\Delta t/2$. Unlike the explicit three-stage leapfrog update above, each ADI half step first updates $\mathbf{E}$ implicitly over $\Delta t/2$, then updates $\mathbf{H}$ explicitly over $\Delta t/2$ from the updated electric field. The first half step treats one set of implicit directions; the second half step treats the complementary directions to complete the cycle. Define the half-step update coefficients

$$C_a = \frac{4\varepsilon - \sigma\Delta t}{4\varepsilon + \sigma\Delta t}, \qquad C_b = \frac{2\Delta t}{4\varepsilon + \sigma\Delta t}, \qquad D_b = \frac{\Delta t}{2\mu}, \tag{3.1}$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity. These differ from the full-step coefficients in (2.1) because each electric update uses $\Delta t/2$.

3.1 First ADI Half-Step

E_x and H_z By equation (1.2a),

$$\varepsilon_{i+\frac{1}{2},j,k}\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - E^{n}_{x,i+\frac{1}{2},j,k}}{\Delta t/2} + \sigma_{i+\frac{1}{2},j,k}\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + E^{n}_{x,i+\frac{1}{2},j,k}}{2} = [H_1], \tag{3.2}$$

where

$$[H_1] = \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}. \tag{3.3}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_x^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_x^{n} + [H_1]. \tag{3.4}$$

Hence

$$E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k}\, [H_1], \tag{3.5}$$

By equation (1.3c),

$$\begin{aligned} H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \Bigg(& \frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} \\ & - \frac{E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta x}\Bigg), \end{aligned} \tag{3.6}$$

Substituting the magnetic update into (3.5) and eliminating $H_z^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} &= C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} \\ &\quad + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \Bigg[H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta x} \right) \\ &\qquad - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k} - D_{b,i+\frac{1}{2},j-\frac{1}{2},k} \left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k}}{\Delta y} - \frac{E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}}{\Delta x} \right) \Bigg] \\ &\quad - \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}} \right). \end{aligned} \tag{3.7}$$

Collecting the unknown $E_x^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} & - \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}\, E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} \\ & + \left[1 + \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} + \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2} \right] E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ & - \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}\, E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} \\ = {} & C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k} \right) \\ & - \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y\, \Delta x} \left(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k} \right) \\ & + \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y\, \Delta x} \left(E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k} \right) \\ & - \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}} \right). \end{aligned} \tag{3.8}$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} + \beta_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - \gamma_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} = p_{x1} E^{n}_{x,i+\frac{1}{2},j,k} \\
+ \frac{1}{\Delta y}\Big(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\Big) \\
- \frac{1}{\Delta z}\Big(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\Big) \\
+ \frac{q_{x1}}{\Delta x \Delta y}\Big(E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}\Big) \\
- \frac{r_{x1}}{\Delta x \Delta y}\Big(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\Big),
\end{aligned}
\tag{3.9}
$$

and the magnetic update can be written as

$$
H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + \frac{r_{x1}}{\Delta y}\Big(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\Big) - \frac{r_{x1}}{\Delta x}\Big(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\Big). \tag{3.10}
$$

where

$$
\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \tag{3.11}
$$

$$
\beta_{x1} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{3.12}
$$

$$
\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \tag{3.13}
$$

$$
p_{x1} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.14}
$$

$$
q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{3.15}
$$

$$
r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.16}
$$

E_y and H_x By equation (1.2b),

$$
\varepsilon_{i,j+\frac{1}{2},k} \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k} \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} + E^{n}_{y,i,j+\frac{1}{2},k}}{2} = [H_2], \tag{3.17}
$$

where

$$
[H_2] = \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}. \tag{3.18}
$$

Rearranging gives

$$
\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_y^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_y^{n} + [H_2]. \tag{3.19}
$$

Hence

$$
E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} = C_{a,i,j+\frac{1}{2},k} E^{n}_{y,i,j+\frac{1}{2},k} + C_{b,i,j+\frac{1}{2},k} [H_2], \tag{3.20}
$$

By equation (1.3a),

$$
\begin{aligned}
H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \Big(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} \\
- \frac{E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta y} \Big),
\end{aligned}
\tag{3.21}
$$

Substituting the magnetic update into (3.20) and eliminating $H_x^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} &= C_{a,i,j+\frac{1}{2},k} E^{n}_{y,i,j+\frac{1}{2},k} \\
&\quad + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\Bigg[H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}\left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} - \frac{E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta y}\right) \\
&\qquad - H^{n}_{x,i,j+\frac{1}{2},k-\frac{1}{2}} - D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}\left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1}}{\Delta z} - \frac{E^{n}_{z,i,j+1,k-\frac{1}{2}} - E^{n}_{z,i,j,k-\frac{1}{2}}}{\Delta y}\right)\Bigg] \\
&\quad - \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x}\left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right).
\end{aligned}
\tag{3.22}
$$

Collecting the unknown $E_y^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} \\
&+ \left[1 + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2}\right] E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \\
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1} \\
= {} & C_{a,i,j+\frac{1}{2},k} E^{n}_{y,i,j+\frac{1}{2},k} + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\left(H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z\, \Delta y}\left(E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}\right) \\
&+ \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z\, \Delta y}\left(E^{n}_{z,i,j+1,k-\frac{1}{2}} - E^{n}_{z,i,j,k-\frac{1}{2}}\right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x}\left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right).
\end{aligned}
\tag{3.23}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1} + \beta_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - \gamma_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} = {} & p_{y1} E^{n}_{y,i,j+\frac{1}{2},k} \\
&+ \frac{1}{\Delta z}\left(H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right) \\
&- \frac{1}{\Delta x}\left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right) \\
&+ \frac{q_{y1}}{\Delta y \Delta z}\left(E^{n}_{z,i,j+1,k-\frac{1}{2}} - E^{n}_{z,i,j,k-\frac{1}{2}}\right) \\
&- \frac{r_{y1}}{\Delta y \Delta z}\left(E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}\right),
\end{aligned}
\tag{3.24}
$$

and the magnetic update can be written as

$$
H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + \frac{r_{y1}}{\Delta z}\left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\right) - \frac{r_{y1}}{\Delta y}\left(E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}\right). \tag{3.25}
$$

where

$$\alpha_{y1} = \frac{D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2}, \tag{3.26}$$

$$\beta_{y1} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y1} + \gamma_{y1}, \tag{3.27}$$

$$\gamma_{y1} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2}, \tag{3.28}$$

$$p_{y1} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{3.29}$$

$$q_{y1} = D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}, \tag{3.30}$$

$$r_{y1} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{3.31}$$

E_z and H_y By equation (1.2c),

$$\varepsilon_{i,j,k+\frac{1}{2}} \frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}}{\Delta t/2} + \sigma_{i,j,k+\frac{1}{2}} \frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} + E^n_{z,i,j,k+\frac{1}{2}}}{2} = [H_3], \tag{3.32}$$

where

$$[H_3] = \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}. \tag{3.33}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_z^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_z^n + [H_3]. \tag{3.34}$$

Hence

$$E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} = C_{a,i,j,k+\frac{1}{2}}\, E^n_{z,i,j,k+\frac{1}{2}} + C_{b,i,j,k+\frac{1}{2}}\, [H_3], \tag{3.35}$$

By equation (1.3b),

$$\begin{aligned} H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \Big(& \frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} \\ & - \frac{E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta z} \Big), \end{aligned} \tag{3.36}$$

Substituting the magnetic update into (3.35) and eliminating $H_y^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} &= C_{a,i,j,k+\frac{1}{2}}\, E^n_{z,i,j,k+\frac{1}{2}} \\ &\quad + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \Bigg[H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta z} \right) \\ &\qquad - H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}} - D_{b,i-\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^n_{x,i-\frac{1}{2},j,k+1} - E^n_{x,i-\frac{1}{2},j,k}}{\Delta z} \right) \Bigg] \\ &\quad - \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}} \right). \end{aligned} \tag{3.37}$$

Collecting the unknown $E_z^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
&-\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}\,E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}}\\
&+\left[1+\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}+\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}\right]E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\\
&-\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}\,E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}}\\
=\,&C_{a,i,j,k+\frac{1}{2}}\,E^n_{z,i,j,k+\frac{1}{2}}+\frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x}\left(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}}-H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right)\\
&-\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x\,\Delta z}\left(E^n_{x,i+\frac{1}{2},j,k+1}-E^n_{x,i+\frac{1}{2},j,k}\right)\\
&+\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x\,\Delta z}\left(E^n_{x,i-\frac{1}{2},j,k+1}-E^n_{x,i-\frac{1}{2},j,k}\right)\\
&-\frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y}\left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}}-H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\right).
\end{aligned}
\tag{3.38}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{z1}E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}}+\beta_{z1}E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}-\gamma_{z1}E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}}=\,&p_{z1}E^n_{z,i,j,k+\frac{1}{2}}\\
&+\frac{1}{\Delta x}\left(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}}-H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right)\\
&-\frac{1}{\Delta y}\left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}}-H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\right)\\
&+\frac{q_{z1}}{\Delta z\Delta x}\left(E^n_{x,i-\frac{1}{2},j,k+1}-E^n_{x,i-\frac{1}{2},j,k}\right)\\
&-\frac{r_{z1}}{\Delta z\Delta x}\left(E^n_{x,i+\frac{1}{2},j,k+1}-E^n_{x,i+\frac{1}{2},j,k}\right),
\end{aligned}
\tag{3.39}
$$

and the magnetic update can be written as

$$
H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}}=H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}}+\frac{r_{z1}}{\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}}-E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right)-\frac{r_{z1}}{\Delta z}\left(E^n_{x,i+\frac{1}{2},j,k+1}-E^n_{x,i+\frac{1}{2},j,k}\right). \tag{3.40}
$$

where

$$\alpha_{z1}=\frac{D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.41}$$

$$\beta_{z1}=\frac{1}{C_{b,i,j,k+\frac{1}{2}}}+\alpha_{z1}+\gamma_{z1}, \tag{3.42}$$

$$\gamma_{z1}=\frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.43}$$

$$p_{z1}=\frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{3.44}$$

$$q_{z1}=D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}, \tag{3.45}$$

$$r_{z1}=D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{3.46}$$

In the AMReX Yee grid, with vacuum media, the field updates are implemented as

$$
\begin{aligned}
-E_x^{n+\frac{1}{2}}(i,j-1,k) + \left(2 + \frac{4\Delta y^2}{c^2\Delta t^2}\right) E_x^{n+\frac{1}{2}}(i,j,k) - E_x^{n+\frac{1}{2}}(i,j+1,k) &= \frac{4\Delta y^2}{c^2\Delta t^2} E_x^n(i,j,k) \\
&+ \frac{2\Delta y^2}{\Delta t}\left[\frac{B_z^n(i,j,k) - B_z^n(i,j-1,k)}{\Delta y} - \frac{B_y^n(i,j,k) - B_y^n(i,j,k-1)}{\Delta z}\right] \\
&+ \Delta y \left[\frac{E_y^n(i+1,j-1,k) - E_y^n(i,j-1,k)}{\Delta x} - \frac{E_y^n(i+1,j,k) - E_y^n(i,j,k)}{\Delta x}\right],
\end{aligned} \tag{3.47a}
$$

$$
\begin{aligned}
-E_y^{n+\frac{1}{2}}(i,j,k-1) + \left(2 + \frac{4\Delta z^2}{c^2\Delta t^2}\right) E_y^{n+\frac{1}{2}}(i,j,k) - E_y^{n+\frac{1}{2}}(i,j,k+1) &= \frac{4\Delta z^2}{c^2\Delta t^2} E_y^n(i,j,k) \\
&+ \frac{2\Delta z^2}{\Delta t}\left[\frac{B_x^n(i,j,k) - B_x^n(i,j,k-1)}{\Delta z} - \frac{B_z^n(i,j,k) - B_z^n(i-1,j,k)}{\Delta x}\right] \\
&+ \Delta z \left[\frac{E_z^n(i,j+1,k-1) - E_z^n(i,j,k-1)}{\Delta y} - \frac{E_z^n(i,j+1,k) - E_z^n(i,j,k)}{\Delta y}\right],
\end{aligned} \tag{3.47b}
$$

$$
\begin{aligned}
-E_z^{n+\frac{1}{2}}(i-1,j,k) + \left(2 + \frac{4\Delta x^2}{c^2\Delta t^2}\right) E_z^{n+\frac{1}{2}}(i,j,k) - E_z^{n+\frac{1}{2}}(i+1,j,k) &= \frac{4\Delta x^2}{c^2\Delta t^2} E_z^n(i,j,k) \\
&+ \frac{2\Delta x^2}{\Delta t}\left[\frac{B_y^n(i,j,k) - B_y^n(i-1,j,k)}{\Delta x} - \frac{B_x^n(i,j,k) - B_x^n(i,j-1,k)}{\Delta y}\right] \\
&+ \Delta x \left[\frac{E_x^n(i-1,j,k+1) - E_x^n(i-1,j,k)}{\Delta z} - \frac{E_x^n(i,j,k+1) - E_x^n(i,j,k)}{\Delta z}\right],
\end{aligned} \tag{3.47c}
$$

$$
B_x^{n+\frac{1}{2}}(i,j,k) = B_x^n(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_y^{n+\frac{1}{2}}(i,j,k+1) - E_y^{n+\frac{1}{2}}(i,j,k)}{\Delta z} - \frac{E_z^n(i,j+1,k) - E_z^n(i,j,k)}{\Delta y}\right], \tag{3.47d}
$$

$$
B_y^{n+\frac{1}{2}}(i,j,k) = B_y^n(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_z^{n+\frac{1}{2}}(i+1,j,k) - E_z^{n+\frac{1}{2}}(i,j,k)}{\Delta x} - \frac{E_x^n(i,j,k+1) - E_x^n(i,j,k)}{\Delta z}\right], \tag{3.47e}
$$

$$
B_z^{n+\frac{1}{2}}(i,j,k) = B_z^n(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_x^{n+\frac{1}{2}}(i,j+1,k) - E_x^{n+\frac{1}{2}}(i,j,k)}{\Delta y} - \frac{E_y^n(i+1,j,k) - E_y^n(i,j,k)}{\Delta x}\right]. \tag{3.47f}
$$

3.2 Second ADI Half-Step

E_x and H_y By equation (1.2a),

$$
\varepsilon_{i+\frac{1}{2},j,k} \frac{E^{n+1}_{x,i+\frac{1}{2},j,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta t/2} + \sigma_{i+\frac{1}{2},j,k} \frac{E^{n+1}_{x,i+\frac{1}{2},j,k} + E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{2} = [H_4], \tag{3.48}
$$

where

$$
[H_4] = \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+1}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}. \tag{3.49}
$$

Rearranging gives

$$
\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_x^{n+1} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_x^{n+\frac{1}{2}} + [H_4]. \tag{3.50}
$$

Hence

$$
E^{n+1}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k} [H_4], \tag{3.51}
$$

By equation (1.3b),

$$H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \Bigg(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta z} \Bigg), \tag{3.52}$$

Substituting the magnetic update into (3.51) and eliminating H_y^{n+1} terms gives

$$\begin{aligned} E^{n+1}_{x,i+\frac{1}{2},j,k} &= C_{a,i+\frac{1}{2},j,k} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ &\quad + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k} \right) \\ &\quad - \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \Bigg[H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta z} \right) \\ &\qquad - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}} - D_{b,i+\frac{1}{2},j,k-\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k-\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k-\frac{1}{2}}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j,k} - E^{n+1}_{x,i+\frac{1}{2},j,k-1}}{\Delta z} \right) \Bigg]. \end{aligned} \tag{3.53}$$

Collecting the unknown E_x^{n+1} terms gives

$$\begin{aligned} &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2} E^{n+1}_{x,i+\frac{1}{2},j,k+1} \\ &+ \left[1 + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2} \right] E^{n+1}_{x,i+\frac{1}{2},j,k} \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2} E^{n+1}_{x,i+\frac{1}{2},j,k-1} \\ = {} & C_{a,i+\frac{1}{2},j,k} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k} \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}} \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z\, \Delta x} \left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \right) \\ &+ \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z\, \Delta x} \left(E^{n+\frac{1}{2}}_{z,i+1,j,k-\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k-\frac{1}{2}} \right). \end{aligned} \tag{3.54}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned} -\alpha_{x2} E^{n+1}_{x,i+\frac{1}{2},j,k-1} + \beta_{x2} E^{n+1}_{x,i+\frac{1}{2},j,k} - \gamma_{x2} E^{n+1}_{x,i+\frac{1}{2},j,k+1} = {} & p_{x2} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ &+ \frac{1}{\Delta y} \left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k} \right) \\ &- \frac{1}{\Delta z} \left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}} \right) \\ &+ \frac{q_{x2}}{\Delta x \Delta z} \left(E^{n+\frac{1}{2}}_{z,i+1,j,k-\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k-\frac{1}{2}} \right) \\ &- \frac{r_{x2}}{\Delta x \Delta z} \left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \right), \end{aligned} \tag{3.55}$$

and the magnetic update can be written as

$$H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + \frac{r_{x2}}{\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right) - \frac{r_{x2}}{\Delta z}\left(E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}\right). \tag{3.56}$$

where

$$\alpha_{x2} = \frac{D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2}, \tag{3.57}$$

$$\beta_{x2} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x2} + \gamma_{x2}, \tag{3.58}$$

$$\gamma_{x2} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2}, \tag{3.59}$$

$$p_{x2} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.60}$$

$$q_{x2} = D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}, \tag{3.61}$$

$$r_{x2} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{3.62}$$

E_y **and** H_z By equation (1.2b),

$$\varepsilon_{i,j+\frac{1}{2},k}\frac{E^{n+1}_{y,i,j+\frac{1}{2},k} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k}\frac{E^{n+1}_{y,i,j+\frac{1}{2},k} + E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{2} = [H_5], \tag{3.63}$$

where

$$[H_5] = \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+1}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}. \tag{3.64}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E^{n+1}_y = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E^{n+\frac{1}{2}}_y + [H_5]. \tag{3.65}$$

Hence

$$E^{n+1}_{y,i,j+\frac{1}{2},k} = C_{a,i,j+\frac{1}{2},k} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} + C_{b,i,j+\frac{1}{2},k}[H_5], \tag{3.66}$$

By equation (1.3c),

$$H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k}\Bigg(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n+1}_{y,i+1,j+\frac{1}{2},k} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta x}\Bigg), \tag{3.67}$$

Substituting the magnetic update into (3.66) and eliminating H^{n+1}_z terms gives

$$\begin{aligned}
E^{n+1}_{y,i,j+\frac{1}{2},k} &= C_{a,i,j+\frac{1}{2},k} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \\
&\quad + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right) \\
&\quad - \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x}\Bigg[H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k}\left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n+1}_{y,i+1,j+\frac{1}{2},k} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta x}\right) \\
&\qquad - H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k} - D_{b,i-\frac{1}{2},j+\frac{1}{2},k}\left(\frac{E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n+1}_{y,i,j+\frac{1}{2},k} - E^{n+1}_{y,i-1,j+\frac{1}{2},k}}{\Delta x}\right)\Bigg].
\end{aligned} \tag{3.68}$$

Collecting the unknown E_y^{n+1} terms gives

$$
\begin{aligned}
&-\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}\,E^{n+1}_{y,i+1,j+\frac{1}{2},k}\\
&+\left[1+\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}+\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}\right]E^{n+1}_{y,i,j+\frac{1}{2},k}\\
&-\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}\,E^{n+1}_{y,i-1,j+\frac{1}{2},k}\\
={}&C_{a,i,j+\frac{1}{2},k}\,E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}+\frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}}-H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right)\\
&-\frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k}-H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right)\\
&-\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x\,\Delta y}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right)\\
&+\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x\,\Delta y}\left(E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j,k}\right).
\end{aligned}
\tag{3.69}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{y2}E^{n+1}_{y,i-1,j+\frac{1}{2},k}+\beta_{y2}E^{n+1}_{y,i,j+\frac{1}{2},k}-\gamma_{y2}E^{n+1}_{y,i+1,j+\frac{1}{2},k}={}&p_{y2}E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\\
&+\frac{1}{\Delta z}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}}-H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right)\\
&-\frac{1}{\Delta x}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k}-H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right)\\
&+\frac{q_{y2}}{\Delta y\Delta x}\left(E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j,k}\right)\\
&-\frac{r_{y2}}{\Delta y\Delta x}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right),
\end{aligned}
\tag{3.70}
$$

and the magnetic update can be written as

$$
H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k}=H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k}+\frac{r_{y2}}{\Delta y}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right)-\frac{r_{y2}}{\Delta x}\left(E^{n+1}_{y,i+1,j+\frac{1}{2},k}-E^{n+1}_{y,i,j+\frac{1}{2},k}\right).
\tag{3.71}
$$

where

$$\alpha_{y2}=\frac{D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2},\tag{3.72}$$

$$\beta_{y2}=\frac{1}{C_{b,i,j+\frac{1}{2},k}}+\alpha_{y2}+\gamma_{y2},\tag{3.73}$$

$$\gamma_{y2}=\frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2},\tag{3.74}$$

$$p_{y2}=\frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}},\tag{3.75}$$

$$q_{y2}=D_{b,i-\frac{1}{2},j+\frac{1}{2},k},\tag{3.76}$$

$$r_{y2}=D_{b,i+\frac{1}{2},j+\frac{1}{2},k}.\tag{3.77}$$

E_z and H_x By equation (1.2c),

$$
\varepsilon_{i,j,k+\frac{1}{2}}\frac{E^{n+1}_{z,i,j,k+\frac{1}{2}}-E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta t/2}+\sigma_{i,j,k+\frac{1}{2}}\frac{E^{n+1}_{z,i,j,k+\frac{1}{2}}+E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{2}=[H_6],
\tag{3.78}
$$

where

$$[H_6] = \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+1}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}. \tag{3.79}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_z^{n+1} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_z^{n+\frac{1}{2}} + [H_6]. \tag{3.80}$$

Hence

$$E^{n+1}_{z,i,j,k+\frac{1}{2}} = C_{a,i,j,k+\frac{1}{2}}\, E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} + C_{b,i,j,k+\frac{1}{2}}\, [H_6], \tag{3.81}$$

By equation (1.3a),

$$\begin{aligned} H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \Bigg(&\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} \\ &- \frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y}\Bigg), \end{aligned} \tag{3.82}$$

Substituting the magnetic update into (3.81) and eliminating H_x^{n+1} terms gives

$$\begin{aligned} E^{n+1}_{z,i,j,k+\frac{1}{2}} = {}& C_{a,i,j,k+\frac{1}{2}}\, E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \\ &+ \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x}\left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y}\Bigg[H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}\left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} - \frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y}\right) \\ &\qquad - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}} - D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}\left(\frac{E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k}}{\Delta z} - \frac{E^{n+1}_{z,i,j,k+\frac{1}{2}} - E^{n+1}_{z,i,j-1,k+\frac{1}{2}}}{\Delta y}\right)\Bigg]. \end{aligned} \tag{3.83}$$

Collecting the unknown E_z^{n+1} terms gives

$$\begin{aligned} &- \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}\, E^{n+1}_{z,i,j+1,k+\frac{1}{2}} \\ &+ \left[1 + \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} + \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}\right] E^{n+1}_{z,i,j,k+\frac{1}{2}} \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}\, E^{n+1}_{z,i,j-1,k+\frac{1}{2}} \\ = {}& C_{a,i,j,k+\frac{1}{2}}\, E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x}\left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\right) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y\, \Delta z}\left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\right) \\ &+ \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y\, \Delta z}\left(E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k}\right). \end{aligned} \tag{3.84}$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{z2} E^{n+1}_{z,i,j-1,k+\frac{1}{2}} + \beta_{z2} E^{n+1}_{z,i,j,k+\frac{1}{2}} - \gamma_{z2} E^{n+1}_{z,i,j+1,k+\frac{1}{2}} = {} & p_{z2} E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \\
& + \frac{1}{\Delta x}\left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right) \\
& - \frac{1}{\Delta y}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\right) \qquad (3.85) \\
& + \frac{q_{z2}}{\Delta z \Delta y}\left(E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k}\right) \\
& - \frac{r_{z2}}{\Delta z \Delta y}\left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\right),
\end{aligned}
$$

and the magnetic update can be written as

$$
H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + \frac{r_{z2}}{\Delta z}\left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\right) - \frac{r_{z2}}{\Delta y}\left(E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}\right). \qquad (3.86)
$$

where

$$\alpha_{z2} = \frac{D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}, \qquad (3.87)$$

$$\beta_{z2} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z2} + \gamma_{z2}, \qquad (3.88)$$

$$\gamma_{z2} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}, \qquad (3.89)$$

$$p_{z2} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \qquad (3.90)$$

$$q_{z2} = D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}, \qquad (3.91)$$

$$r_{z2} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \qquad (3.92)$$

In the AMReX Yee grid, with vacuum media, the field updates are implemented as

$$
\begin{aligned}
-E^{n+1}_x(i,j,k-1) + \left(2 + \frac{4\Delta z^2}{c^2\Delta t^2}\right) E^{n+1}_x(i,j,k) - E^{n+1}_x(i,j,k+1) = {} & \frac{4\Delta z^2}{c^2\Delta t^2} E^{n+\frac{1}{2}}_x(i,j,k) \\
& + \frac{2\Delta z^2}{\Delta t}\left[\frac{B^{n+\frac{1}{2}}_z(i,j,k) - B^{n+\frac{1}{2}}_z(i,j-1,k)}{\Delta y} - \frac{B^{n+\frac{1}{2}}_y(i,j,k) - B^{n+\frac{1}{2}}_y(i,j,k-1)}{\Delta z}\right] \\
& + \Delta z\left[\frac{E^{n+\frac{1}{2}}_z(i+1,j,k-1) - E^{n+\frac{1}{2}}_z(i,j,k-1)}{\Delta x} - \frac{E^{n+\frac{1}{2}}_z(i+1,j,k) - E^{n+\frac{1}{2}}_z(i,j,k)}{\Delta x}\right],
\end{aligned}
\qquad (3.93a)
$$

$$
\begin{aligned}
-E^{n+1}_y(i-1,j,k) + \left(2 + \frac{4\Delta x^2}{c^2\Delta t^2}\right) E^{n+1}_y(i,j,k) - E^{n+1}_y(i+1,j,k) = {} & \frac{4\Delta x^2}{c^2\Delta t^2} E^{n+\frac{1}{2}}_y(i,j,k) \\
& + \frac{2\Delta x^2}{\Delta t}\left[\frac{B^{n+\frac{1}{2}}_x(i,j,k) - B^{n+\frac{1}{2}}_x(i,j,k-1)}{\Delta z} - \frac{B^{n+\frac{1}{2}}_z(i,j,k) - B^{n+\frac{1}{2}}_z(i-1,j,k)}{\Delta x}\right] \\
& + \Delta x\left[\frac{E^{n+\frac{1}{2}}_x(i-1,j+1,k) - E^{n+\frac{1}{2}}_x(i-1,j,k)}{\Delta y} - \frac{E^{n+\frac{1}{2}}_x(i,j+1,k) - E^{n+\frac{1}{2}}_x(i,j,k)}{\Delta y}\right],
\end{aligned}
\qquad (3.93b)
$$

$$-E_z^{n+1}(i,j-1,k) + \left(2 + \frac{4\Delta y^2}{c^2\Delta t^2}\right) E_z^{n+1}(i,j,k) - E_z^{n+1}(i,j+1,k) = \frac{4\Delta y^2}{c^2\Delta t^2} E_z^{n+\frac{1}{2}}(i,j,k)$$
$$+ \frac{2\Delta y^2}{\Delta t}\left[\frac{B_y^{n+\frac{1}{2}}(i,j,k) - B_y^{n+\frac{1}{2}}(i-1,j,k)}{\Delta x} - \frac{B_x^{n+\frac{1}{2}}(i,j,k) - B_x^{n+\frac{1}{2}}(i,j-1,k)}{\Delta y}\right]$$
$$+ \Delta y\left[\frac{E_y^{n+\frac{1}{2}}(i,j-1,k+1) - E_y^{n+\frac{1}{2}}(i,j-1,k)}{\Delta z} - \frac{E_y^{n+\frac{1}{2}}(i,j,k+1) - E_y^{n+\frac{1}{2}}(i,j,k)}{\Delta z}\right], \tag{3.93c}$$

$$B_x^{n+1}(i,j,k) = B_x^{n+\frac{1}{2}}(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_y^{n+\frac{1}{2}}(i,j,k+1) - E_y^{n+\frac{1}{2}}(i,j,k)}{\Delta z} - \frac{E_z^{n+1}(i,j+1,k) - E_z^{n+1}(i,j,k)}{\Delta y}\right], \tag{3.93d}$$

$$B_y^{n+1}(i,j,k) = B_y^{n+\frac{1}{2}}(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_z^{n+\frac{1}{2}}(i+1,j,k) - E_z^{n+\frac{1}{2}}(i,j,k)}{\Delta x} - \frac{E_x^{n+1}(i,j,k+1) - E_x^{n+1}(i,j,k)}{\Delta z}\right], \tag{3.93e}$$

$$B_z^{n+1}(i,j,k) = B_z^{n+\frac{1}{2}}(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_x^{n+\frac{1}{2}}(i,j+1,k) - E_x^{n+\frac{1}{2}}(i,j,k)}{\Delta y} - \frac{E_y^{n+1}(i+1,j,k) - E_y^{n+1}(i,j,k)}{\Delta x}\right]. \tag{3.93f}$$

4 Solution of Linear Systems

See Tridiagonal matrix algorithm for more details.

5 Perfect Electric Conductor Boundary Conditions

For a perfect electric conductor (PEC) boundary with outward unit normal $\hat{\mathbf{n}}$, the electric field satisfies

$$\hat{\mathbf{n}} \times \mathbf{E} = 0. \tag{5.1}$$

That is, the tangential electric field vanishes on the conducting surface. If the field inside the conductor is not evolved and the initial data are consistent with the boundary, then Faraday's law also implies that the normal magnetic flux remains zero on the wall,

$$\hat{\mathbf{n}} \cdot \mathbf{B} = 0. \tag{5.2}$$

Now consider a rectangular domain with PEC boundaries on the two faces normal to x, namely the planes $x = x_{\text{lo}}$ and $x = x_{\text{hi}}$, while the four faces normal to y and z remain periodic. On the AMReX Yee grid, E_y and E_z are nodal in x and therefore live directly on the x-boundary planes, whereas E_x is centered at $i + \frac{1}{2}$ in the x direction and is not stored on the boundary itself. If the domain has N_x cells in x, the discrete PEC conditions are therefore

$$E_y(0,j,k) = E_y(N_x,j,k) = 0, \qquad E_z(0,j,k) = E_z(N_x,j,k) = 0. \tag{5.3}$$

No value is imposed directly on E_x at the wall, since it is the normal component.
In the Yee update, the implementation is to overwrite the tangential electric fields on the two PEC planes after each electric-field stage,

$$E_y^m(0,j,k) = E_y^m(N_x,j,k) = 0, \qquad E_z^m(0,j,k) = E_z^m(N_x,j,k) = 0, \tag{5.4}$$

where $m = n + \frac{1}{2}$ or $n+1$ depending on the stage. The neighboring magnetic updates then use differences against these zero boundary values. For example, the x-derivative appearing in the B_y update is approximated by

$$\left.\frac{\partial E_z}{\partial x}\right|_{i+\frac{1}{2},j,k} \approx \frac{E_z(i+1,j,k) - E_z(i,j,k)}{\Delta x}. \tag{5.5}$$

At the first interior face next to $x = x_{\text{lo}}$,

$$\left.\frac{\partial E_z}{\partial x}\right|_{\frac{1}{2},j,k} \approx \frac{E_z(1,j,k) - E_z(0,j,k)}{\Delta x} = \frac{E_z(1,j,k)}{\Delta x}, \tag{5.6}$$

while at the last interior face next to $x = x_{\text{hi}}$,

$$\left.\frac{\partial E_z}{\partial x}\right|_{N_x-\frac{1}{2},j,k} \approx \frac{E_z(N_x,j,k) - E_z(N_x-1,j,k)}{\Delta x} = -\frac{E_z(N_x-1,j,k)}{\Delta x}. \tag{5.7}$$

Likewise, the update for the normal magnetic component B_x on the PEC planes involves only tangential derivatives of E_y and E_z, so if B_x is initialized consistently on the wall then it remains zero there.
In the ADI scheme, the same PEC condition enters as a Dirichlet condition whenever a tangential electric component is solved implicitly along x. Because the y and z directions remain periodic, the implicit solves along those directions are unchanged and stay cyclic. The only modified line solves are the x-directed ones:

- the E_z solve in the first half-step, equation (3.39);
- the E_y solve in the second half-step, equation (3.70).

For the first half-step, the E_z solve uses

$$E^{n+\frac{1}{2}}_{z,0,j,k+\frac{1}{2}} = 0, \qquad E^{n+\frac{1}{2}}_{z,N_x,j,k+\frac{1}{2}} = 0. \tag{5.8}$$

Hence the first and last interior tridiagonal equations are

$$\beta_{z1} E^{n+\frac{1}{2}}_{z,1,j,k+\frac{1}{2}} - \gamma_{z1} E^{n+\frac{1}{2}}_{z,2,j,k+\frac{1}{2}} = \mathrm{RHS}, \tag{5.9}$$

and

$$-\alpha_{z1} E^{n+\frac{1}{2}}_{z,N_x-2,j,k+\frac{1}{2}} + \beta_{z1} E^{n+\frac{1}{2}}_{z,N_x-1,j,k+\frac{1}{2}} = \mathrm{RHS}, \tag{5.10}$$

respectively. The boundary unknowns do not participate in the solve; they are prescribed by the PEC condition.
Similarly, in the second half-step the E_y solve uses

$$E^{n+1}_{y,0,j+\frac{1}{2},k} = 0, \qquad E^{n+1}_{y,N_x,j+\frac{1}{2},k} = 0. \tag{5.11}$$

Hence the first and last interior tridiagonal equations are

$$\beta_{y2} E^{n+1}_{y,1,j+\frac{1}{2},k} - \gamma_{y2} E^{n+1}_{y,2,j+\frac{1}{2},k} = \mathrm{RHS}, \tag{5.12}$$

and

$$-\alpha_{y2} E^{n+1}_{y,N_x-2,j+\frac{1}{2},k} + \beta_{y2} E^{n+1}_{y,N_x-1,j+\frac{1}{2},k} = \mathrm{RHS}, \tag{5.13}$$

respectively.
Note that we do not need to explicitly impose $\hat{\mathbf{n}} \cdot \mathbf{B} = 0$ separately. For an x-normal wall, the magnetic update on the boundary plane $i = i_b$, with $i_b \in \{0, N_x\}$, is

$$B_x^{n+1}(i_b,j,k) = B_x^n(i_b,j,k) + \frac{\Delta t}{2}\left[\frac{E_y^{n+\frac{1}{2}}(i_b,j,k+1) - E_y^{n+\frac{1}{2}}(i_b,j,k)}{\Delta z} - \frac{E_z^{n+1}(i_b,j+1,k) - E_z^{n+1}(i_b,j,k)}{\Delta y}\right]. \tag{5.14}$$

Since the PEC condition gives

$$E_y(i_b,j,k) = 0, \qquad E_z(i_b,j,k) = 0 \tag{5.15}$$

for all j and k, it follows that

$$B_x^{n+1}(i_b,j,k) = B_x^n(i_b,j,k). \tag{5.16}$$

Therefore, if

$$B_x^0(i_b,j,k) = 0, \tag{5.17}$$

then

$$B_x^n(i_b,j,k) = 0 \tag{5.18}$$

is already satisfied for all n.

6 Dispersion and PEC Normal-Mode Analysis

This section addresses two related but distinct questions. First, on an infinite or periodic grid, we prescribe a Fourier wavenumber and determine the temporal eigenvalue; this gives the bulk numerical dispersion relation. Second, on a half-space terminated by a PEC, we prescribe a possible growing temporal eigenvalue and any tangential Fourier wavenumber, determine the admissible spatial roots normal to the wall, and test whether their linear combination can satisfy the PEC condition. The second calculation is the discrete boundary normal-mode analogue of a Laplace–Fourier half-plane analysis.
We separate the geometries because the boundary-normal coordinate cannot be Fourier-transformed. The calculations below proceed through three model problems:

1. the bulk periodic problem, used to derive dispersion;
2. an x-normal PEC with no tangential variation, used as a one-dimensional boundary warm-up;
3. a y-normal PEC with Fourier modes in both tangential directions, used for the full three-dimensional Maxwell boundary problem.

In each half-space problem the stability question is whether a nontrivial mode can be both temporally growing, $|\lambda| > 1$, and spatially bounded away from the wall.
Begin with a homogeneous-vacuum wave propagating in the x-direction with

$$E_x = E_y = 0, \qquad H_x = H_z = 0, \tag{6.1}$$

so that the only nontrivial fields are E_z and H_y. The other axis-aligned polarizations follow by a cyclic permutation of coordinates. For simplicity, we consider the case of vacuum media. Set $\sigma = 0$, $\varepsilon = \varepsilon_0$, $\mu = \mu_0$, and

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}. \tag{6.2}$$

The half-step coefficients (3.1) are then

$$C_a = 1, \qquad C_b = \frac{\Delta t}{2\varepsilon_0}, \qquad D_b = \frac{\Delta t}{2\mu_0}. \tag{6.3}$$

Define the coefficient γ by

$$\gamma := C_b D_b = \frac{(c\Delta t)^2}{4}. \tag{6.4}$$

Define the two staggered first-difference operators by

$$\left(\delta_x^E E_z\right)_{i+\frac{1}{2},j,k+\frac{1}{2}} := \frac{E_{z,i+1,j,k+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}}{\Delta x}, \tag{6.5}$$

$$\left(\delta_x^H H_y\right)_{i,j,k+\frac{1}{2}} := \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x}. \tag{6.6}$$

Thus δ_x^E maps the E_z grid to the H_y grid, while δ_x^H maps the H_y grid to the E_z grid. Their composition is the centered second difference

$$\begin{aligned} \left(\delta_x^2 E_z\right)_{i,j,k+\frac{1}{2}} &:= \left(\delta_x^H \delta_x^E E_z\right)_{i,j,k+\frac{1}{2}} \\ &= \frac{E_{z,i-1,j,k+\frac{1}{2}} - 2E_{z,i,j,k+\frac{1}{2}} + E_{z,i+1,j,k+\frac{1}{2}}}{\Delta x^2}. \end{aligned} \tag{6.7}$$

6.1 One-dimensional ADI update

Restricting (3.39)–(3.40) to this polarization and multiplying the electric equation by C_b, the first half-step is

$$\left(1 - \gamma\delta_x^2\right) E_z^{n+\frac{1}{2}} = E_z^n + C_b \delta_x^H H_y^n, \tag{6.8}$$

$$H_y^{n+\frac{1}{2}} = H_y^n + D_b \delta_x^E E_z^{n+\frac{1}{2}}, \tag{6.9}$$

where E_z is implicit along x. In the second half-step, E_z is formally implicit along y. Because this mode has no y-dependence, that second-difference term vanishes and the update reduces to

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b \delta_x^H H_y^{n+\frac{1}{2}}, \tag{6.10}$$

$$H_y^{n+1} = H_y^{n+\frac{1}{2}} + D_b \delta_x^E E_z^{n+\frac{1}{2}}. \tag{6.11}$$

Both magnetic half-updates use the same intermediate electric field; therefore,

$$H_y^{n+1} = H_y^n + 2D_b \delta_x^E E_z^{n+\frac{1}{2}} \tag{6.12}$$

6.2 Plane-wave ansatz and Yee symbols

At any time level m, write a spatial Fourier mode on the staggered Yee grid as

$$\begin{aligned} E^m_{z,i,j,k+\frac{1}{2}} &= \widehat{E}_z^{\,m} e^{\mathrm{i}k_x i\Delta x}, \\ H^m_{y,i+\frac{1}{2},j,k+\frac{1}{2}} &= \widehat{H}_y^{\,m} e^{\mathrm{i}k_x (i+\frac{1}{2})\Delta x}. \end{aligned} \tag{6.13}$$

Plugging into (6.5) gives

$$\begin{aligned} \left(\delta_x^E E_z^m\right)_{i+\frac{1}{2},j,k+\frac{1}{2}} &= \widehat{E}_z^{\,m} e^{\mathrm{i}k_x(i+\frac{1}{2})\Delta x} \frac{e^{\mathrm{i}k_x\Delta x/2} - e^{-\mathrm{i}k_x\Delta x/2}}{\Delta x} \\ &= \mathrm{i}K_x \widehat{E}_z^{\,m} e^{\mathrm{i}k_x(i+\frac{1}{2})\Delta x}, \end{aligned} \tag{6.14}$$

where

$$K_x := \frac{2}{\Delta x}\sin\left(\frac{k_x\Delta x}{2}\right). \tag{6.15}$$

Similarly, (6.6) gives

$$\begin{aligned} \left(\delta_x^H H_y^m\right)_{i,j,k+\frac{1}{2}} &= \widehat{H}_y^{\,m} e^{\mathrm{i}k_x i\Delta x} \frac{e^{\mathrm{i}k_x\Delta x/2} - e^{-\mathrm{i}k_x\Delta x/2}}{\Delta x} \\ &= \mathrm{i}K_x \widehat{H}_y^{\,m} e^{\mathrm{i}k_x i\Delta x}. \end{aligned} \tag{6.16}$$

Consequently, the Fourier symbols of the explicitly defined operators are

$$\delta_x^E \longleftrightarrow \mathrm{i}K_x, \qquad \delta_x^H \longleftrightarrow \mathrm{i}K_x, \qquad \delta_x^2 \longleftrightarrow -K_x^2. \tag{6.17}$$

As $\Delta x \to 0$, $K_x \to k_x$, so these symbols recover their continuum counterparts.

6.3 Fourier reduction of the half-steps

For brevity, define the derivative symbol and implicit denominator

$$s := \mathrm{i}K_x, \qquad D := 1 + \gamma K_x^2. \tag{6.18}$$

Applying the symbols (6.17) to (6.8), the first-half electric update becomes

$$D\widehat{E}_z^{\,n+\frac{1}{2}} = \widehat{E}_z^{\,n} + C_b s \widehat{H}_y^{\,n}. \tag{6.19}$$

Applying the same symbols to (6.9) gives

$$\widehat{H}_y^{\,n+\frac{1}{2}} = \widehat{H}_y^{\,n} + D_b s \widehat{E}_z^{\,n+\frac{1}{2}}. \tag{6.20}$$

Likewise, (6.10) becomes

$$\widehat{E}_z^{\,n+1} = \widehat{E}_z^{\,n+\frac{1}{2}} + C_b s \widehat{H}_y^{\,n+\frac{1}{2}}, \tag{6.21}$$

Finally, combining the two magnetic half-steps as in (6.11) gives

$$\widehat{H}_y^{\,n+1} = \widehat{H}_y^{\,n} + 2D_b s \widehat{E}_z^{\,n+\frac{1}{2}} \tag{6.22}$$

6.4 Amplification matrix

Solving (6.19) for the intermediate electric amplitude gives

$$\widehat{E}_z^{\,n+\frac{1}{2}} = \frac{1}{D}\widehat{E}_z^{\,n} + \frac{C_b s}{D}\widehat{H}_y^{\,n}. \tag{6.23}$$

Using (6.20) in (6.21), and then substituting the intermediate electric amplitude above, gives

$$\begin{aligned} \widehat{E}_z^{\,n+1} &= \frac{1-\gamma K_x^2}{D}\widehat{E}_z^{\,n} + \frac{2C_b s}{D}\widehat{H}_y^{\,n}, \\ \widehat{H}_y^{\,n+1} &= \frac{2D_b s}{D}\widehat{E}_z^{\,n} + \frac{1-\gamma K_x^2}{D}\widehat{H}_y^{\,n}, \end{aligned} \tag{6.24}$$

where $C_b D_b = \gamma$, $s^2 = -K_x^2$, and $D = 1 + \gamma K_x^2$ have been used. The one-step map is therefore

$$\begin{pmatrix} \widehat{E}_z \\ \widehat{H}_y \end{pmatrix}^{n+1} = G \begin{pmatrix} \widehat{E}_z \\ \widehat{H}_y \end{pmatrix}^{n}, \qquad G = \frac{1}{D}\begin{pmatrix} 1-\gamma K_x^2 & 2C_b s \\ 2D_b s & 1-\gamma K_x^2 \end{pmatrix}. \tag{6.25}$$

6.5 Numerical dispersion relation

A time-harmonic mode satisfies

$$\begin{pmatrix} \widehat{E}_z \\ \widehat{H}_y \end{pmatrix}^{n+1} = \lambda \begin{pmatrix} \widehat{E}_z \\ \widehat{H}_y \end{pmatrix}^{n}, \qquad \lambda = e^{-\mathrm{i}\omega\Delta t}. \tag{6.26}$$

Using (6.25), the characteristic equation can be written as

$$\begin{aligned} 0 = D^2 \det(G - \lambda I) &= \left(1 - \gamma K_x^2 - \lambda D\right)^2 - 4 C_b D_b s^2 \\ &= \left(1 - \gamma K_x^2 - \lambda D\right)^2 + 4\gamma K_x^2. \end{aligned} \tag{6.27}$$

Solving this quadratic for λ yields

$$\lambda = \frac{1 - \gamma K_x^2 \pm 2\mathrm{i}\, K_x \sqrt{\gamma}}{1 + \gamma K_x^2}. \tag{6.28}$$

Introduce the CFL number $S := c\Delta t/\Delta x$ and

$$\nu := S \sin\left(\frac{k_x \Delta x}{2}\right) = K_x \sqrt{\gamma}. \tag{6.29}$$

Then $\gamma K_x^2 = \nu^2$, and the eigenvalues simplify to

$$\lambda = \frac{1 \pm \mathrm{i}\nu}{1 \mp \mathrm{i}\nu}. \tag{6.30}$$

The forward-propagating branch ($\omega > 0$) is

$$\lambda = \frac{1 - \mathrm{i}\nu}{1 + \mathrm{i}\nu}, \tag{6.31}$$

for which $|\lambda| = 1$ (unconditional stability of this one-dimensional reduction) and

$$\arg(\lambda) = -2\arctan(\nu). \tag{6.32}$$

Therefore the ADI numerical dispersion relation is

$$\boxed{\omega_{\mathrm{ADI}}(k_x) = \frac{2}{\Delta t}\,\arctan\left(S \sin\frac{k_x \Delta x}{2}\right).} \tag{6.33}$$

In the continuum limit $k_x\Delta x \to 0$ with fixed S, one has $\sin(k_x\Delta x/2) \approx k_x\Delta x/2$ and $\arctan\nu \approx \nu$, hence

$$\omega_{\mathrm{ADI}} \approx \frac{2}{\Delta t} \cdot S \cdot \frac{k_x \Delta x}{2} = c k_x, \tag{6.34}$$

as required. The leading truncation error is $O((k_x\Delta x)^2)$ at fixed CFL number, so the phase-velocity error decreases quadratically with cells-per-wavelength.

6.6 Normal-mode analysis of the PEC boundary

We now apply the one-dimensional update (6.8)–(6.12) to the x-normal PEC described in Section 5. Consider the left boundary as a half-space problem, $i \geq 0$. For the same E_z–H_y polarization used above, (5.3) reduces to

$$E_{z,0}^m = 0 \tag{6.35}$$

at every electric-field stage m.

In the bulk ansatz (6.13), a shift by one cell in x has eigenvalue $e^{\mathrm{i}k_x\Delta x}$, whose modulus is one. A boundary mode must instead allow an unknown complex eigenvalue ρ for this shift. Retaining the temporal factor λ^n from (6.26) and the same Yee staggering as (6.13), replace $e^{\mathrm{i}k_x\Delta x}$ by ρ and write

$$\begin{aligned} E_{z,i}^n &= \widehat{E}_z \lambda^n \rho^i, \\ H_{y,i+\frac{1}{2}}^n &= \widehat{H}_y \lambda^n \rho^{i+\frac{1}{2}}. \end{aligned} \tag{6.36}$$

Advancing the mode by one full time step multiplies both fields by λ, so $|\lambda| > 1$ corresponds to exponential growth in time. Likewise, increasing i by one in (6.36) gives

$$\frac{E_{z,i+1}^n}{E_{z,i}^n} = \rho, \qquad \frac{H_{y,i+\frac{3}{2}}^n}{H_{y,i+\frac{1}{2}}^n} = \rho. \tag{6.37}$$

Thus each one-cell shift away from the PEC multiplies either field by ρ, and after i shifts its magnitude contains the factor $|\rho|^i$. Hence boundedness as $i \to +\infty$ requires $|\rho| \le 1$, while $|\rho| < 1$ gives exponential decay into the interior and therefore a mode localized near the PEC.

Apply the electric-to-magnetic staggered difference (6.5) to the first line of (6.36):

$$\begin{aligned}\left(\delta_x^E E_z^n\right)_{i+\frac{1}{2}} &= \frac{\widehat{E}_z \lambda^n \rho^{i+1} - \widehat{E}_z \lambda^n \rho^i}{\Delta x} \\ &= \widehat{E}_z \lambda^n \rho^{i+\frac{1}{2}} q(\rho),\end{aligned} \tag{6.38}$$

where

$$q(\rho) := \frac{\rho^{1/2} - \rho^{-1/2}}{\Delta x}. \tag{6.39}$$

Applying the magnetic-to-electric difference (6.6) gives the same factor:

$$\begin{aligned}\left(\delta_x^H H_y^n\right)_i &= \frac{\widehat{H}_y \lambda^n \rho^{i+\frac{1}{2}} - \widehat{H}_y \lambda^n \rho^{i-\frac{1}{2}}}{\Delta x} \\ &= \widehat{H}_y \lambda^n \rho^i q(\rho).\end{aligned} \tag{6.40}$$

Composing the two staggered first differences consequently multiplies the electric mode by $q(\rho)^2$.[1] Thus the Fourier symbols (6.17) are replaced by

$$\delta_x^E \longleftrightarrow q(\rho), \qquad \delta_x^H \longleftrightarrow q(\rho), \qquad \delta_x^2 \longleftrightarrow q(\rho)^2. \tag{6.41}$$

Write $q := q(\rho)$. Applying the normal-mode symbols to the first-half electric update (6.8) gives

$$\left(1 - \gamma q^2\right) \widehat{E}_z^{n+\frac{1}{2}} = \widehat{E}_z^n + C_b q \widehat{H}_y^n. \tag{6.42}$$

The first-half magnetic update (6.9) becomes

$$\widehat{H}_y^{n+\frac{1}{2}} = \widehat{H}_y^n + D_b q \widehat{E}_z^{n+\frac{1}{2}}. \tag{6.43}$$

Likewise, the second-half electric update (6.10) becomes

$$\widehat{E}_z^{n+1} = \widehat{E}_z^{n+\frac{1}{2}} + C_b q \widehat{H}_y^{n+\frac{1}{2}}, \tag{6.44}$$

and combining the two magnetic half-steps as in (6.12) gives

$$\widehat{H}_y^{n+1} = \widehat{H}_y^n + 2 D_b q \widehat{E}_z^{n+\frac{1}{2}}. \tag{6.45}$$

Equations (6.42)–(6.45) are the normal-mode counterparts of (6.19)–(6.22): $\mathrm{i}K_x$ is replaced by q, and $1 + \gamma K_x^2$ is replaced by $1 - \gamma q^2$. Eliminating the intermediate amplitudes exactly as in the amplification-matrix derivation gives

$$G(q) = \frac{1}{1 - \gamma q^2} \begin{pmatrix} 1 + \gamma q^2 & 2 C_b q \\ 2 D_b q & 1 + \gamma q^2 \end{pmatrix}. \tag{6.46}$$

The temporal normal-mode condition is

$$G(q) \begin{pmatrix} \widehat{E}_z \\ \widehat{H}_y \end{pmatrix} = \lambda \begin{pmatrix} \widehat{E}_z \\ \widehat{H}_y \end{pmatrix}. \tag{6.47}$$

Its characteristic equation is

$$0 = \left(1 - \gamma q^2\right)^2 \det(G(q) - \lambda I) = \left[1 + \gamma q^2 - \lambda \left(1 - \gamma q^2\right)\right]^2 - 4 \gamma q^2. \tag{6.48}$$

Factoring the resulting quadratic yields

$$\lambda = \frac{1 \pm \sqrt{\gamma}\, q}{1 \mp \sqrt{\gamma}\, q}, \tag{6.49}$$

or, equivalently,

$$\sqrt{\gamma}\, q = \pm \frac{\lambda - 1}{\lambda + 1}. \tag{6.50}$$

[1] For a bulk Fourier mode, $\rho = e^{\mathrm{i}k_x \Delta x}$, and (6.39) gives $q(\rho) = \left(e^{\mathrm{i}k_x \Delta x/2} - e^{-\mathrm{i}k_x \Delta x/2}\right)/\Delta x = \mathrm{i}K_x$ by (6.15). Thus the preceding Fourier analysis is the unit-circle case $|\rho| = 1$ of the normal-mode construction.

For $|\lambda| > 1$,

$$\Re\left(\frac{\lambda - 1}{\lambda + 1}\right) = \frac{|\lambda|^2 - 1}{|\lambda + 1|^2} > 0, \tag{6.51}$$

so the two values of q in (6.50) have real parts of opposite sign.
To recover the actual grid mode, set $z = \rho^{1/2}$. Equation (6.39) gives

$$z^2 - q\Delta x\, z - 1 = 0. \tag{6.52}$$

The two roots of (6.52) satisfy $z_1 z_2 = -1$. Since $\rho_j = z_j^2$, their corresponding spatial factors satisfy $\rho_1\rho_2 = 1$. Moreover, neither root lies on the unit circle when $|\lambda| > 1$: if $\rho = e^{\mathrm{i}\theta}$, then (6.39) makes q purely imaginary, contradicting $\Re q \neq 0$ from (6.50). Thus exactly one root ρ_- satisfies $|\rho_-| < 1$ and is retained by boundedness; its reciprocal is rejected.
Let

$$\Phi_- := \begin{pmatrix} \widehat{E}_z \\ \rho_-^{1/2}\widehat{H}_y \end{pmatrix} \tag{6.53}$$

be the staggered interior eigenvector associated with ρ_-. The general bounded mode contains one undetermined coefficient,

$$U_i^n = \theta\lambda^n \rho_-^i \Phi_-. \tag{6.54}$$

The PEC boundary operator extracts the electric component,

$$\mathcal{B}_x := \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad K_{\mathrm{PEC}}^{(x)}(\lambda) := \mathcal{B}_x \Phi_-. \tag{6.55}$$

Substitution into (6.35) produces the scalar boundary system

$$K_{\mathrm{PEC}}^{(x)}(\lambda)\theta = 0. \tag{6.56}$$

For $|\lambda| > 1$, $q \neq 0$ and the coupled eigenproblem (6.46) has $\widehat{E}_z \neq 0$. We may therefore normalize $\widehat{E}_z = 1$, for which $K_{\mathrm{PEC}}^{(x)} = [1]$ and

$$\det K_{\mathrm{PEC}}^{(x)}(\lambda) = 1 \neq 0. \tag{6.57}$$

Hence $\theta = 0$: the one-dimensional half-space has no nontrivial bounded mode with $|\lambda| > 1$.

7 External Excitation

External field excitations are prescribed by user-defined grid functions of space and time, together with a spatial flag that selects the excitation type at each Yee edge (or face). Following the Artemis convention,

- flag = 0: no excitation,
- flag = 1: hard source (Dirichlet), field equals the prescribed value,
- flag = 2: soft source, the prescribed value is *added* to the field update.

In the macroscopic ADI push these two families are treated differently: soft electric sources are built into the implicit electric systems, while magnetic sources are applied after each magnetic half-step. Hard electric sources are not supported in the ADI electric solve.

7.1 Soft Electric Sources in the ADI Electric Update

Write $S^{n+\frac{1}{2}}_{E_x,i+\frac{1}{2},j,k}$ for the soft-source value assigned to the E_x edge at $\left(i+\frac{1}{2},j,k\right)$ and the arrival time level $n+\frac{1}{2}$ (and likewise for E_y and E_z at their staggered locations). Over a full step the intended soft contribution at that edge is split equally between the two ADI half-steps and evaluated at the resulting field levels $n+\frac{1}{2}$ and $n+1$. The first-half electric update (3.5) is then modified to

$$E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k}\,[H_1] + \frac{1}{2}\, S^{n+\frac{1}{2}}_{E_x,i+\frac{1}{2},j,k}, \tag{7.1}$$

on edges where the E_x flag equals 2, and is unchanged elsewhere. The second half-step (3.51) is analogous,

$$E^{n+1}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k}\, E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k}\,[H_4] + \frac{1}{2}\, S^{n+1}_{E_x,i+\frac{1}{2},j,k}. \tag{7.2}$$

The same construction applies componentwise to E_y and E_z. The magnetic updates themselves are unchanged by the electric source; only the electric right-hand side is affected.

7.2 Effect on the ADI Tridiagonal System

Dividing (7.1) by $C_{b,i+\frac{1}{2},j,k}$ and substituting the magnetic update as in Section 3 yields the same left-hand side as (3.9). On soft-source edges the only modification is an additional known term on the right-hand side:

$$
\begin{aligned}
-\alpha_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} + \beta_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - \gamma_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} = p_{x1}E^{n}_{x,i+\frac{1}{2},j,k} \\
+ \frac{1}{\Delta y}\Big(H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^n_{z,i+\frac{1}{2},j-\frac{1}{2},k}\Big) \\
- \frac{1}{\Delta z}\Big(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\Big) \\
+ \frac{q_{x1}}{\Delta x\Delta y}\Big(E^n_{y,i+1,j-\frac{1}{2},k} - E^n_{y,i,j-\frac{1}{2},k}\Big) \\
- \frac{r_{x1}}{\Delta x\Delta y}\Big(E^n_{y,i+1,j+\frac{1}{2},k} - E^n_{y,i,j+\frac{1}{2},k}\Big) \\
+ \frac{1}{2C_{b,i+\frac{1}{2},j,k}}\,S^{n+\frac{1}{2}}_{E_x,i+\frac{1}{2},j,k},
\end{aligned}
\tag{7.3}
$$

with the same coefficients $\alpha_{x1}, \beta_{x1}, \gamma_{x1}, p_{x1}, q_{x1}, r_{x1}$ as in Section 3. Away from soft-source edges the final term is omitted. In the implementation this contribution is the increment

$$
\texttt{rhs}_{i,j,k} \mathrel{+}= \tfrac{1}{2}\,S^{n+\frac{1}{2}}_{E_x,i+\frac{1}{2},j,k}/C_{b,i+\frac{1}{2},j,k}
\tag{7.4}
$$

added after the curl terms have been assembled. The E_y and E_z first-half systems are modified in exactly the same way, and the second half-step is analogous with $S^{n+1}_{E_x,i+\frac{1}{2},j,k}/(2C_{b,i+\frac{1}{2},j,k})$ on the normalized right-hand side.

Consequently a hard electric source (flag = 1), which would require constraining the unknown E itself rather than adding to the right-hand side, is rejected in the ADI electric path.

7.3 Magnetic Sources After Each Half-Step

After the first ADI half-step has produced $\mathbf{E}^{n+\frac{1}{2}}$ and $\mathbf{B}^{n+\frac{1}{2}}$, and again after the second half-step has produced $\mathbf{E}^{n+1}$ and $\mathbf{B}^{n+1}$, an external magnetic excitation is applied directly to the magnetic field. The excitation is evaluated at the arrival level of the field being modified. Thus, after the first half-step, a soft magnetic source (flag = 2) contributes

$$
B_{x,i,j+\frac{1}{2},k+\frac{1}{2}} \leftarrow B_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + \tfrac{1}{2}\,S^{n+\frac{1}{2}}_{B_x,i,j+\frac{1}{2},k+\frac{1}{2}},
\tag{7.5}
$$

while a hard magnetic source (flag = 1) overwrites

$$
B_{x,i,j+\frac{1}{2},k+\frac{1}{2}} \leftarrow S^{n+\frac{1}{2}}_{B_x,i,j+\frac{1}{2},k+\frac{1}{2}}.
\tag{7.6}
$$

After the second half-step the corresponding source values are $S^{n+1}_{B_x}$: a soft source adds $\frac{1}{2}S^{n+1}_{B_x}$, while a hard source overwrites the field with $S^{n+1}_{B_x}$. (The factor $\frac{1}{2}$ again splits a full-step soft increment across the two half-steps.) The same rules apply to B_y and B_z at their staggered faces. Unlike the electric soft source, this magnetic update is an explicit post-processing of the field after the ADI magnetic stage; it does not enter the tridiagonal electric systems.

7.4 Summary of the ADI Excitation Path

Over one full step Δt, the macroscopic ADI scheme therefore proceeds as follows.

1. Form and solve the first-half electric systems with soft electric sources $S_E^{n+\frac{1}{2}}$ added as $\frac{1}{2}S_E^{n+\frac{1}{2}}$ (equivalently $\frac{1}{2}S_E^{n+\frac{1}{2}}/C_b$ on the normalized RHS).
2. Update $\mathbf{B}$ to $n+\frac{1}{2}$ from the ADI magnetic formulae.
3. Apply magnetic excitation to $\mathbf{B}^{n+\frac{1}{2}}$ (soft: add $\frac{1}{2}S_B^{n+\frac{1}{2}}$; hard: set equal to $S_B^{n+\frac{1}{2}}$).
4. Form and solve the second-half electric systems with soft electric sources S_E^{n+1}.
5. Update $\mathbf{B}$ to $n+1$, then apply magnetic excitation evaluated at $n+1$.

Hard electric sources are not part of this path. Soft electric sources never overwrite E; they only augment the Ampère right-hand side of each half-step, consistently with the half-step coefficients C_a and C_b of (3.1).

8 Lumped Inductance

ADI removes the Maxwell CFL restriction, so a lumped inductor must not reintroduce a local timestep limit from the LC scale $\omega_L \sim \sqrt{\kappa/\varepsilon}$. This section embeds an ideal inductor in the ADI electric update by a centered trapezoidal coupling: it is nondissipative on the isolated field–inductor subsystem and places no restriction on $\omega_L \Delta t$.

8.1 Continuum model

A lumped inductor enters Ampère's law as an additional current density $\mathbf{J}_\mathrm{L}$,

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} + \mathbf{J}_\mathrm{L} \tag{8.1a}$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}. \tag{8.1b}$$

On each Yee edge that hosts an inductor, the constitutive relation $V = L\, dI/dt$ is written in field form. For an x-directed edge of length Δx and face area $\Delta y\, \Delta z$, with total inductance L_x assigned to that edge,

$$V = E_x \Delta x, \qquad I = J_{\mathrm{L},x}\, \Delta y\, \Delta z, \tag{8.2}$$

so

$$E_x \Delta x = L_x\, \Delta y\, \Delta z\, \frac{\partial J_{\mathrm{L},x}}{\partial t}, \tag{8.3}$$

or equivalently

$$\frac{\partial J_{\mathrm{L},x}}{\partial t} = \kappa_x E_x, \qquad \kappa_x = \frac{\Delta x}{\Delta y\, \Delta z\, L_x}. \tag{8.4}$$

The y and z components are analogous, with

$$\kappa_y = \frac{\Delta y}{\Delta x\, \Delta z\, L_y}, \qquad \kappa_z = \frac{\Delta z}{\Delta x\, \Delta y\, L_z}. \tag{8.5}$$

Away from inductor edges one sets $\kappa = 0$ (equivalently $L \to \infty$), so $\mathbf{J}_\mathrm{L}$ need not be stored or evolved there; outside the inductor region it should normally be initialized to zero. Faraday's law (8.1b) is unmodified, so the magnetic ADI updates of Section 3 are unchanged. Only the electric update and the inductor ODE are coupled.

8.2 Centered trapezoidal coupling

Over a half step of size $\Delta t/2$, discretize (8.4) by the trapezoidal rule,

$$\frac{J^{n+\frac{1}{2}}_{\mathrm{L},x,i+\frac{1}{2},j,k} - J^{n}_{\mathrm{L},x,i+\frac{1}{2},j,k}}{\Delta t/2} = \kappa_{x,i+\frac{1}{2},j,k}\, \frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + E^{n}_{x,i+\frac{1}{2},j,k}}{2}. \tag{8.6}$$

Ampère's law must use the same centered average $(J^{n+\frac{1}{2}}_{\mathrm{L},x} + J^{n}_{\mathrm{L},x})/2$; using only the advanced value $J^{n+\frac{1}{2}}_{\mathrm{L},x}$ is dissipative (Section 8.3). With the same conductivity average as in Section 3,

$$\varepsilon_{i+\frac{1}{2},j,k} \frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - E^{n}_{x,i+\frac{1}{2},j,k}}{\Delta t/2} + \sigma_{i+\frac{1}{2},j,k} \frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + E^{n}_{x,i+\frac{1}{2},j,k}}{2} + \frac{J^{n+\frac{1}{2}}_{\mathrm{L},x,i+\frac{1}{2},j,k} + J^{n}_{\mathrm{L},x,i+\frac{1}{2},j,k}}{2} = [H_1], \tag{8.7}$$

where $[H_1]$ is unchanged from Section 3. From (8.6),

$$\frac{J^{n+\frac{1}{2}}_{\mathrm{L},x,i+\frac{1}{2},j,k} + J^{n}_{\mathrm{L},x,i+\frac{1}{2},j,k}}{2} = J^{n}_{\mathrm{L},x,i+\frac{1}{2},j,k} + \frac{\Delta t}{8}\, \kappa_{x,i+\frac{1}{2},j,k} \big(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + E^{n}_{x,i+\frac{1}{2},j,k}\big). \tag{8.8}$$

Substituting into (8.7) and collecting unknowns gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} + \frac{\Delta t}{8}\kappa_x\right) E^{n+\frac{1}{2}}_x = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} - \frac{\Delta t}{8}\kappa_x\right) E^{n}_x + [H_1] - J^{n}_{\mathrm{L},x}. \tag{8.9}$$

Define the inductor-modified half-step coefficients

$$C_a^L = \frac{\dfrac{2\varepsilon}{\Delta t} - \dfrac{\sigma}{2} - \dfrac{\Delta t}{8}\kappa}{\dfrac{2\varepsilon}{\Delta t} + \dfrac{\sigma}{2} + \dfrac{\Delta t}{8}\kappa}, \qquad C_b^L = \frac{1}{\dfrac{2\varepsilon}{\Delta t} + \dfrac{\sigma}{2} + \dfrac{\Delta t}{8}\kappa}. \tag{8.10}$$

When $\kappa = 0$ these reduce to (3.1). The electric update is then

$$E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} = C^L_{a,i+\frac{1}{2},j,k} E^n_{x,i+\frac{1}{2},j,k} + C^L_{b,i+\frac{1}{2},j,k}\Big([H_1] - J^n_{\mathrm{L},x,i+\frac{1}{2},j,k}\Big). \tag{8.11}$$

Thus κ modifies the local diagonal coefficients C_a^L and C_b^L, while the known current J_L^n enters the right-hand side. After $E_x^{n+\frac{1}{2}}$ is obtained, rearrange (**??**) to advance $J_{\mathrm{L},x}$ by

$$J^{n+\frac{1}{2}}_{\mathrm{L},x,i+\frac{1}{2},j,k} = J^n_{\mathrm{L},x,i+\frac{1}{2},j,k} + \frac{\Delta t}{4}\kappa_{x,i+\frac{1}{2},j,k}\big(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + E^n_{x,i+\frac{1}{2},j,k}\big). \tag{8.12}$$

8.3 Local energy and why the current must be averaged

On an inductor edge with $\kappa > 0$, $\sigma = 0$, and no curl contribution, define

$$\mathcal{E} = \frac{\varepsilon}{2}E^2 + \frac{1}{2\kappa}J_\mathrm{L}^2. \tag{8.13}$$

The centered pair

$$\varepsilon\frac{E^{n+\frac{1}{2}} - E^n}{\Delta t/2} + \frac{J_\mathrm{L}^{n+\frac{1}{2}} + J_\mathrm{L}^n}{2} = 0, \qquad \frac{J_\mathrm{L}^{n+\frac{1}{2}} - J_\mathrm{L}^n}{\Delta t/2} = \kappa\frac{E^{n+\frac{1}{2}} + E^n}{2}, \tag{8.14}$$

conserves this energy exactly: $\mathcal{E}^{n+\frac{1}{2}} - \mathcal{E}^n = 0$. Replacing the Ampère current by the fully advanced value $J_\mathrm{L}^{n+\frac{1}{2}}$ instead yields

$$\mathcal{E}^{n+\frac{1}{2}} - \mathcal{E}^n = -\frac{1}{2\kappa}\big(J_\mathrm{L}^{n+\frac{1}{2}} - J_\mathrm{L}^n\big)^2 \le 0, \tag{8.15}$$

so the scheme is numerically dissipative even though the inductor ODE itself is integrated by the trapezoidal rule. That one-sided choice removes the $\omega_L\Delta t$ restriction but acts like an artificial resistance; we reject it for that reason.

8.4 Effect on the ADI tridiagonal system

The magnetic update for $H_z^{n+\frac{1}{2}}$ is identical to Section 3. Substituting it into (8.11) yields the same stencil as (3.9), but with $C_{a,i+\frac{1}{2},j,k} \to C^L_{a,i+\frac{1}{2},j,k}$ and $C_{b,i+\frac{1}{2},j,k} \to C^L_{b,i+\frac{1}{2},j,k}$:

$$\begin{aligned}
-\alpha_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} + \beta^L_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - \gamma_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} = p^L_{x1}&E^n_{x,i+\frac{1}{2},j,k} \\
&+ \frac{1}{\Delta y}\Big(H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^n_{z,i+\frac{1}{2},j-\frac{1}{2},k}\Big) \\
&- \frac{1}{\Delta z}\Big(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\Big) \\
&+ \frac{q_{x1}}{\Delta x\Delta y}\Big(E^n_{y,i+1,j-\frac{1}{2},k} - E^n_{y,i,j-\frac{1}{2},k}\Big) \\
&- \frac{r_{x1}}{\Delta x\Delta y}\Big(E^n_{y,i+1,j+\frac{1}{2},k} - E^n_{y,i,j+\frac{1}{2},k}\Big) \\
&- J^n_{\mathrm{L},x,i+\frac{1}{2},j,k},
\end{aligned} \tag{8.16}$$

where

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \tag{8.17}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \tag{8.18}$$

$$\beta^L_{x1} = \frac{1}{C^L_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{8.19}$$

$$p^L_{x1} = \frac{C^L_{a,i+\frac{1}{2},j,k}}{C^L_{b,i+\frac{1}{2},j,k}}, \tag{8.20}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{8.21}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{8.22}$$

The off-diagonal coefficients α_{x1} and γ_{x1} are unchanged (they come only from the implicit curl of H). The diagonal β^L_{x1} and the factor p^L_{x1} feel L_x through $C^L_{b,i+\frac{1}{2},j,k}$ and $C^L_{a,i+\frac{1}{2},j,k}$. The E_y and E_z first-half systems are modified in the same way.

The current update is

$$J^{n+\frac{1}{2}}_{\mathrm{L},x,i+\frac{1}{2},j,k} = J^{n}_{\mathrm{L},x,i+\frac{1}{2},j,k} + \frac{\Delta t}{4}\kappa_{x,i+\frac{1}{2},j,k}\big(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + E^{n}_{x,i+\frac{1}{2},j,k}\big). \tag{8.23}$$

The second half-step is analogous.

8.5 Stability

Eliminating J_L locally produces an oscillator of frequency $\omega_L = \sqrt{\kappa/\varepsilon}$. On the isolated lossless field–inductor subsystem, the centered scheme of Section 8.3 has amplification factors of unit modulus for every Δt, so it introduces no lumped-inductor CFL. That local fact is not by itself a complete proof for the full spatial ADI splitting; unconditional stability of the coupled scheme requires that the same centered coupling be embedded consistently in an energy-stable ADI discretization.

By contrast, the explicit rule $J^{n+\frac{1}{2}}_\mathrm{L} = J^n_\mathrm{L} + (\Delta t/2)\kappa E^n$, with the known current inserted only as a right-hand-side source, leaves $\omega_L \Delta t$ constrained and defeats the purpose of ADI at large CFL. That is the leapfrog macroscopic FDTD inductor, and it is unsuitable here.

8.6 Present implementation

In the current Artemis macroscopic ADI push, soft electric sources are added to the ADI right-hand side, but $\mathbf{J}_\mathrm{L}$ is not. The inductor object still advances $\mathbf{J}_\mathrm{L}$ explicitly when enabled, yet that current is unused by `MacroscopicEvolveADI`. A stable ADI inductor therefore requires: (i) replace C_a, C_b by C^L_a, C^L_b on inductor edges; (ii) add $-C^L_b J^{\mathrm{old}}_\mathrm{L}$ to the field-update form, or equivalently $-J^{\mathrm{old}}_\mathrm{L}$ to the normalized tridiagonal right-hand side; and (iii) update J_L with (8.12) after each electric solve.